

Mapping Conformational Changes in the Saliva Proteome Potentially Associated with Oral Cancer Aggressiveness

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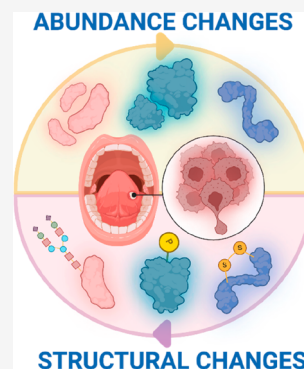
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ABSTRACT: Diverse proteomics-based strategies have been applied to saliva to quantitatively identify diagnostic and prognostic targets for oral cancer. Considering that these targets may be regulated by events that do not imply variation in protein abundance levels, we hypothesized that changes in protein conformation can be associated with diagnosis and prognosis, revealing biological processes and novel targets of clinical relevance. For this, we employed limited proteolysis–mass spectrometry in saliva samples to explore structural alterations, comparing the proteome of healthy control and oral squamous cell carcinoma (OSCC) patients with and without lymph node metastasis. Thirty-six proteins with potential structural rearrangements were associated with clinical patient features including transketolase and its interacting partners. Moreover, *N*-glycosylated peptides contribute to structural rearrangements of potential diagnostic and prognostic markers. Altogether, this approach utilizes saliva proteins to search for targets for diagnosing and prognosing oral cancer and can guide the discovery of potential regulated sites beyond protein-level abundance.

KEYWORDS: *limited proteolysis, mass spectrometry, oral cancer, protein post-translational modification, prognosis, saliva*



INTRODUCTION

Saliva is a promising fluid for biomarker investigation because it reflects pathological and physiological alterations in local and distant parts of the body, representing a more accessible screening tool that can be repeatedly monitored over time.¹ Moreover, the salivary proteome has been characterized in several diseases, including oral squamous cell carcinoma (OSCC),^{2,3} mostly considering the traditional proteomics approach for evaluating protein abundance levels and post-translational modifications (PTMs). Saliva's high complexity in terms of dynamic range and broad composition^{4,5} can lead to great challenges in proteome coverage and require different strategies to interrogate its proteome and detect changes associated with OSCC disease. Thus, investigating protein conformational changes in saliva can help overcome some of these challenges and contribute to the identification of novel diagnostic and prognostic markers for oral cancer in addition to contributing to the identification of functional states and sites.

To this end, we applied the limited proteolysis-mass spectrometry (LiP-MS) strategy to saliva samples from healthy individuals compared to saliva samples from OSCC patients with or without lymph node metastasis. LiP-MS revealed 253 proteins with differential structural rearrangements, here named as conformotypic proteins, compared to 23 differential proteins identified and quantified based on the traditional denaturing

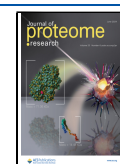
workflow (Trypsin-only). Moreover, our study provides information on uniquely enriched biological processes identified for the conformotypic proteins and indicates 36 of these proteins associated with key clinicopathological features in oral cancer, including alcohol consumption, smoking habit, sample group, status, lymph node metastasis, and tumor size. Finally, we investigated the source of structural rearrangements, indicating potential diagnostic and prognostic proteins with several predicted and experimentally confirmed PTMs, mainly glycosylation, which may help explain the conformational changes or the modifications that modulate the enzymatic cleavage of the targets.

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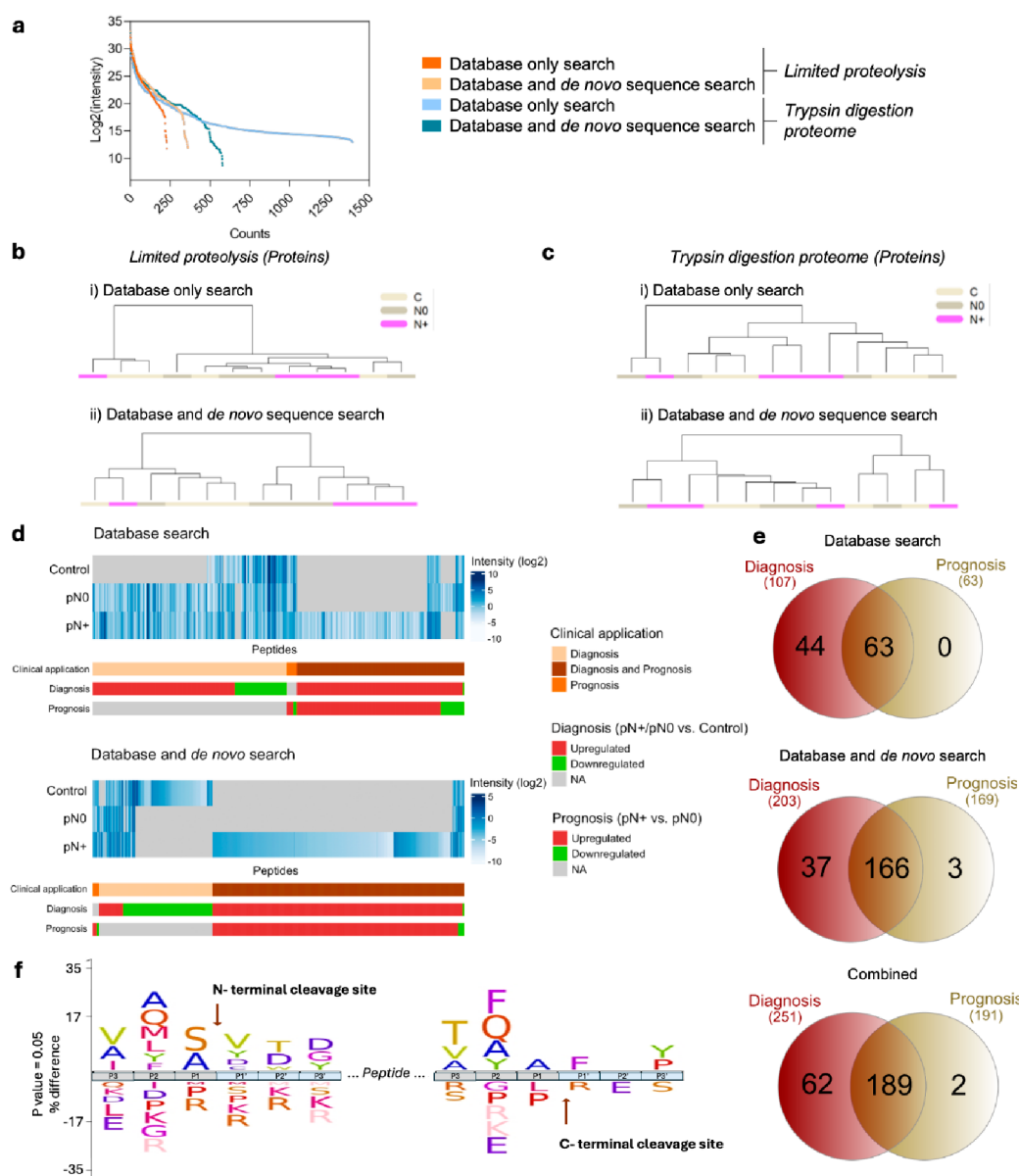


Figure 1. OSCC and lymph node metastasis-associated structural changes revealed by LiP-MS in the human saliva proteome. (a) Dynamic range graphs show the relative distribution of protein abundance counts in LiP-MS and trypsin digestion-only proteome in both search strategies. (b,c) Dendrogram of the 12 LC-MS runs per condition (Limited Proteolysis and Trypsin digestion) searched as “database only” and “database and *de novo*” searches. Clustering analyses by Pearson correlation and Ward linkage were performed using results from proteins identified in LiP-MS and in tryptic digestion-only proteome, indicating partial grouping of biological samples (C/pN0/pN+). (d) Heatmap of peptide intensities associated with diagnosis (pN+ or pN0 vs control; p -value ≤ 0.05 ; ANOVA combined with peptides detected exclusively in one of two sample groups) or prognosis (pN+ vs pN0; p -value ≤ 0.05 ; ANOVA combined with peptides detected exclusively in one of these sample groups) considering database-only search (upper panel; 293 differentially abundant LiP peptides) or database-and-*de novo* search (bottom panel; 637 differentially abundant LiP peptides). (e) Venn diagrams illustrating the identification overlap of source proteins of peptides associated with diagnosis and prognosis using “database-only-search” and/or “database-and-*de novo* sequence search”. NA: not applicable. (f) Icelogo indicates position-specific amino acid residues underrepresented and overrepresented in the putative N and C-terminal cleavage sites of nontryptic and semitryptic LiP differential derived peptides.

RESULTS AND DISCUSSION

LiP-MS Strategies Reveal a Higher Number of Biomarker Targets Compared to Tryptic Digestion-Only Proteome

Classical experiments evaluating the proteome of saliva, the fluid in which the primary oral tumor is embedded, have led to the identification of promising proteins as potential diagnostic and prognostic signatures for OSCC, improving our understanding, and potential management of the disease.^{2–4} However, conformational changes in proteins present in saliva as potential biomarkers have not been previously associated with oral cancer.

In this scenario, saliva has a great advantage for structural analysis since it does not require protein extraction under denaturing conditions; therefore, its native proteome can be directly studied by LiP-MS workflow.^{6,7}

Through LiP-MS combined with a classical proteome workflow, we investigated changes in the abundance and structure of proteins from the saliva of OSCC patients and from control individuals (Figure S1), aiming to identify conformational changes with clinical value. The detailed clinical and pathological information for the eight OSCC patients and

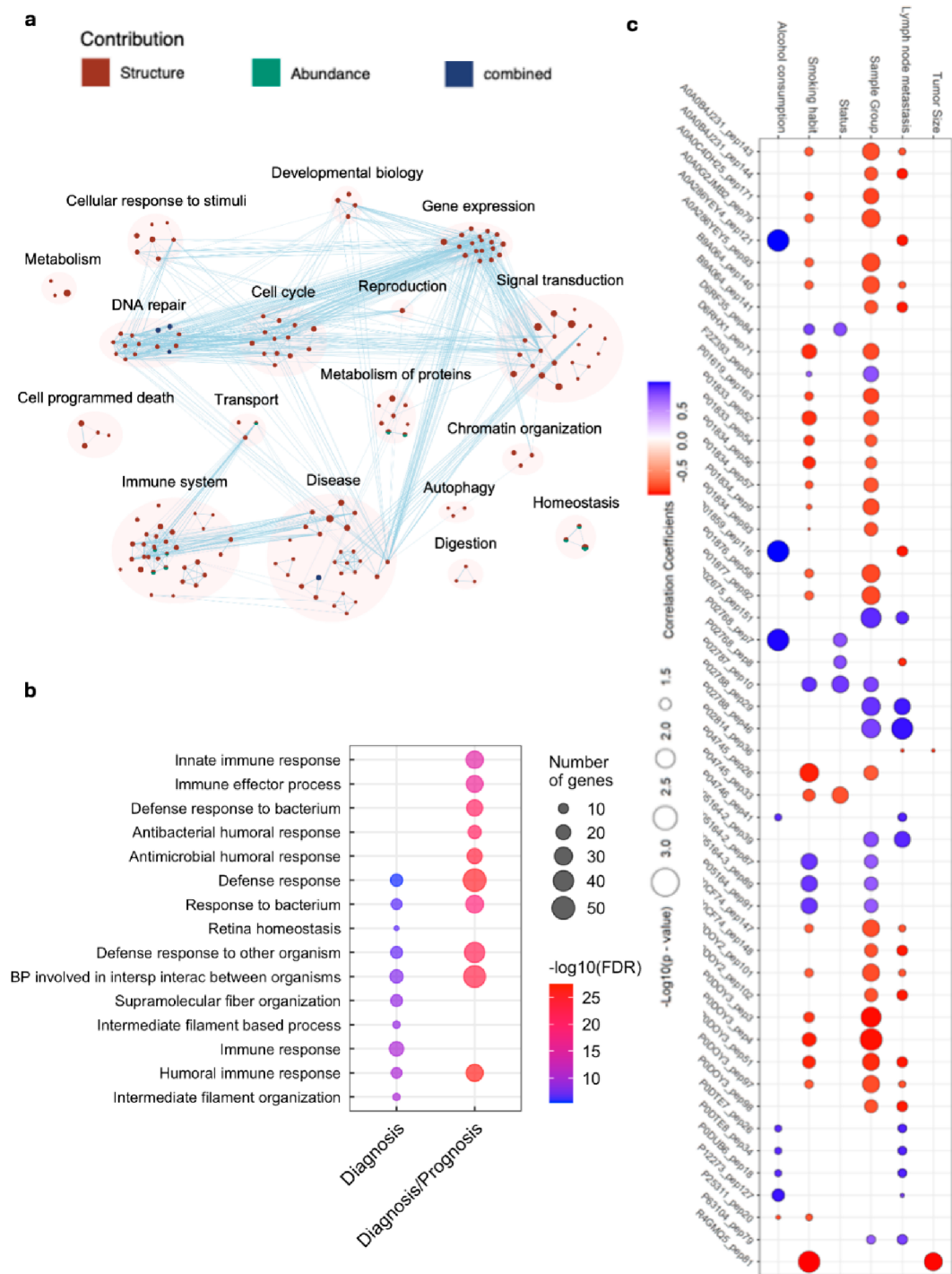


Figure 2. Functional analysis comparing structural (LiP-MS) and protein abundance changes in OSCC. (a) Enrichment map showing REACTOME pathways enriched for proteins containing the differential peptides identified by structural ($n = 253$ proteins) and abundance ($n = 23$ proteins) approaches ($FDR \leq 0.01$). Nodes represent pathways, and the major functional groups are highlighted by filled circles. Enrichment significance (FDR) is represented as node sizes, and pathways with common genes are shown to be connected. Nodes are colored based on supporting evidence from structural or abundance proteomes. Combined evidence was detected through data fusion and not when any of the input data sets were detected separately. (b) Visualization of the top-10 GO biological processes overrepresented for proteins represented in Figure 1e from peptides associated with diagnosis and prognosis ($n = 189$ proteins) or only with diagnosis ($n = 62$ proteins) using GSEA ($FDR \leq 0.05$). (c) LiP-MS-peptides having potential clinical value for OSCC diagnosis and prognosis, indicated by their association with clinical features. Status indicates being alive or not alive. Sample group indicates control, pN0 and pN+.

the clinical information on the four healthy individuals enrolled in this study as well as the mass spectrometry running order are summarized in Table S1 and S2. For this, the output of two different search algorithms (MaxQuant,⁸ PEAKS⁹) was analyzed separately to generate a list of conformatypic peptides, that is, peptides with conformational changes identified by LiP-MS. Our results show a diverse dynamic range of the saliva proteome profiled according to the digestion strategies and search

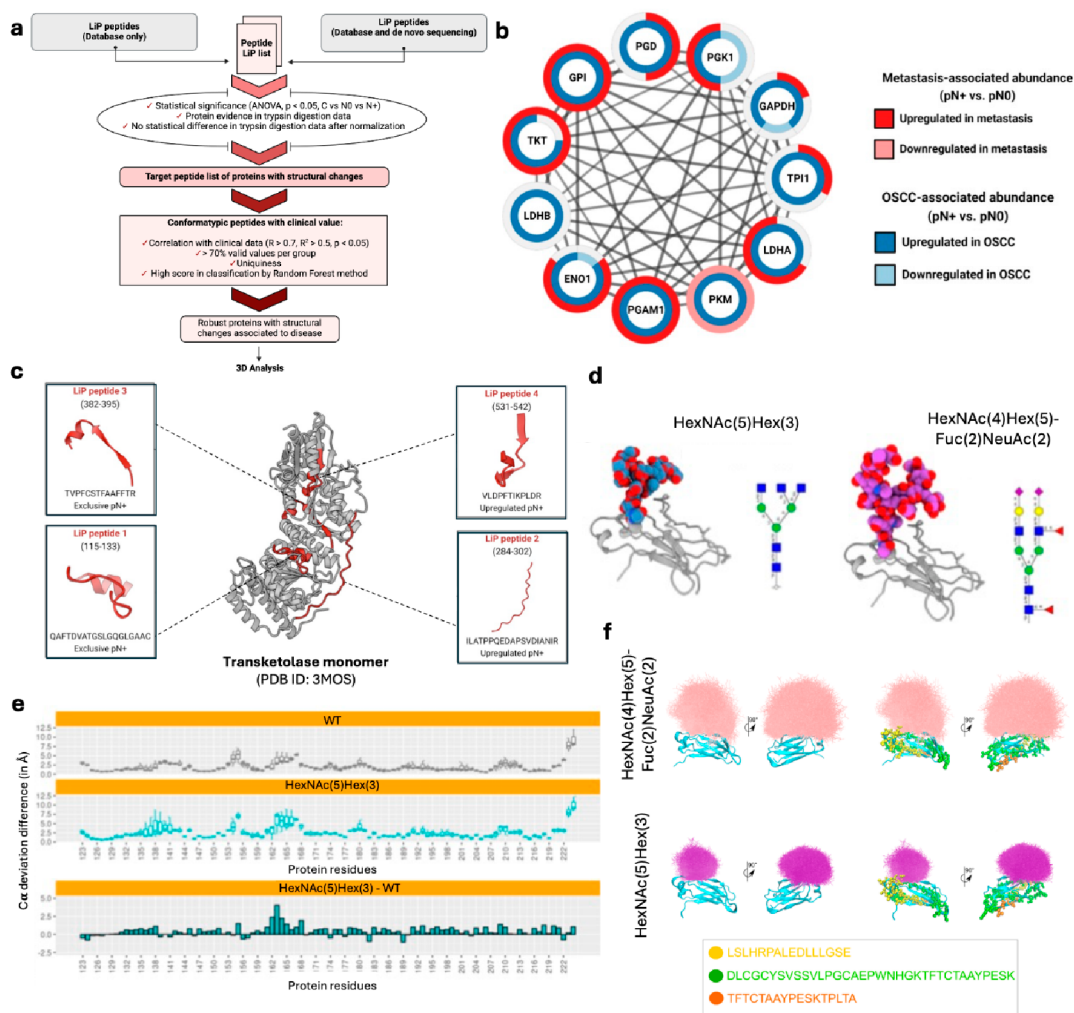


Figure 3. TKT is a strong candidate for conformational changes associated with OSCC. (a) Workflow for LiP target prioritization. (b) Conformotypic peptides are highlighted in red. (c) The STRING network with TKT and functional associated proteins that the LiP-MS strategy indicated structural changes. Each node represents a protein, and edges connecting nodes represent high confidence functional associations (STRING score > 0.7). (d) Three-dimensional models of the glycans HexNAc(5)Hex(3) (in blue, upper model) and HexNAc(4)Hex(5)Fuc(2)NeuAc(2) (in purple, bottom model) bound to the IGHA1 Fc domain through the Asn144 residue. A schematic view of each glycan chain is also presented. (e) Analysis of 100 ns (in triplicate) molecular dynamics simulations in the absence (WT) or presence of the HexNAc(5)Hex(3) glycan. The two upper plots show the distance (only considering C α atoms) between each IGHA1 residue in the simulation trajectory and in a reference initial structure. The reference structure corresponds to the first frame collected from the WT simulation. Higher values mean that the residue deviates from its position in the reference structure during the trajectory. Each boxplot presents the meaning of deviations computed along the simulation trajectory in triplicate. The bottom plot shows the deviation difference per residue upon subtraction of HexNAc(5)Hex(3) glycan distances by WT distances. Higher values mean that a higher deviation was observed in the simulation with IGHA1 bound to HexNAc(5)Hex(3) in comparison to the simulations without glycan. (f) Molecular dynamics simulations of the IGHA1 Fc domain bound to the glycans HexNAc(5)Hex(3) and HexNAc(4)Hex(5)Fuc(2)NeuAc(2). Superposition of the simulation trajectory aligned by the IGHA1 domain (in cyan) and presenting the glycan position (as lines) from every 0.2 ns of the 100 ns trajectory (in triplicate). Only the first frame of the IGHA1 is presented. Pink lines correspond to HexNAc(4) Hex(5)Fuc(2)NeuAc(2) atoms and purple lines correspond to HexNAc(5)Hex(3) atoms. The three conformotypic identified peptides are highlighted as spheres. The sequence of each peptide and its color scheme are shown in the bottom box.

approaches. Providing insights into the heterogeneity of saliva proteome,^{3,4,10,11} unsupervised hierarchical clustering analyses revealed distinct sample grouping of the proteome from tryptic digestion and from LiP-MS, indicating intragroup and intergroup heterogeneity, evidenced by both sample digestion and search approaches (Figure 1a–c, Figure S2, Table S3–S8).

Despite the high intragroup heterogeneity of the saliva proteome, a total of 293 and 637 differential peptides (ANOVA, p -value < 0.05 combined with exclusively detected) were identified in LiP-MS data with MaxQuant and PEAKS tools, respectively, comparing pN0, pN+, and control groups (Figure 1d,e, Table S9–S10), indicating potential diagnostic and

prognostic candidates. To explore potential consensus amino acids in the conformotypic peptides, we performed a motif analysis with the semitryptic and nontryptic differential peptides, which did not reveal a consensus sequence that could indicate conformational changes in preferential residues but highlighted some overrepresented amino acids near the cleavage site (Figure 1f). Interestingly, they account majorly for the residue's alanine, glutamate, and serine mostly hydrophilic residues, that have been previously seen in ordered regions with higher average flexibility (high B-factor).^{12,13}

Furthermore, we investigated the thermal stability of the LiP differential peptides using data from the Meltome Atlas,¹⁴

revealing that these saliva proteins are extremely stable and become insoluble only in high temperatures (Figure S3a, Table S9–S10). These findings could explain the difficulties to observe conformational changes in saliva and mostly in the two conditions investigated in this study (pN+ and pN0), but also demonstrates the efficiency of LiP-MS strategy in identifying those subtle changes.

Exploring the main PTMs identified in or near (± 10 amino acids) the sequence of the differential LiP peptides or proteins, glycosylation was the most common PTM identified in these targets (Figure S3b, Table S11–S17), which may contribute to their higher solubility and could comply with the Meltome Atlas data, which reindicates saliva as a more stable proteome.^{15–17} Also, there is still a complex relationship between PTMs and thermal protein stability^{15,18,19} than previously appreciated that should be further investigated.

Biological Functions and OSCC Clinicopathological Features Propose Relevant LiP-Peptides

Combining the findings from the different search algorithms, we identified 253 proteins structurally altered and with differences in abundance between the patient groups (Figure S4a). Although the LiP-MS strategy indicated a lower dynamic range of the proteome in saliva samples (Figure 1a), these results pointed to a higher number of enriched biological processes and pathways when compared to the traditional proteomics workflow results (Figure 2a,b, Figure S4b, Table S18–S20).

To further explore the potential clinical relevance of our findings in oral cancer (Figure 2c, Table S21), the differential diagnostic and prognostic LiP candidate peptides were evaluated for associations with a range of clinical features. Results indicated potential conformotypic peptides with a significant association with prognostic features (linear regression p -value < 0.05 , and Pearson's R -squared > 0.5), including a subset of 50 peptides correlated with alcohol and smoking consumption, sample grouping, lymph node metastasis and tumor size (Figure 2c, Table S21).

LiP-Diagnostic and Prognostic Peptides Show PTM Based Conformational Changes Performing Patients' Stratification

To explore potential candidates with ability to distinguish OSCC patients, conformotypic peptides were selected if they filled the following criteria: (i) statistical significance (ANOVA, $p < 0.05$), (ii) correlation with clinical data ($R > 0.7$, $R^2 > 0.5$, $p < 0.05$), (iii) discrimination of patient condition. (Figure 3a). Transketolase (TKT), an enzyme that participates in the pentose phosphate pathway, was selected as a potential structural marker in this study. TKT was upregulated in the pN+ condition (from database and de novo search analysis) and showed high performance in discriminating patient conditions by different ranking method analyses (infoGain, SVM, chiSquared) (Figure 3b; Figure S5).

This protein was also previously identified with higher abundance in nonmalignant cells from lymph nodes compared to primary tumor's cells.²⁰ Although TKT has a recognized prognostic role in tumoral tissue,^{21,22} its contribution to saliva has yet to be determined. TKT enzymes are phosphorylated in Thr380 by AKT,²³ and the structural changes detected by LiP-MS encompass this region, implying a role of phosphorylation in TKT conformational change, more specifically in lymph node metastasis condition.²⁴

Evaluating the annotated protein–protein interaction network, TKT shows functional associations with other proteins

identified in this LiP-MS strategy, revealing a network proteome that could also undergo structural changes associated with oral cancer (Figure 3c). To search for structural features that may explain the generation of differential conformotypic peptides in the proteins pointed out in this network, LiP differential peptides were mapped to each protein structure, and different *in silico* analyses were performed. This inspection revealed that in most cases the conformotypic peptides are buried in the structure, not exposed to the solvent, while their N- and C-terminal are mainly secondary structure turns or nonstructured regions exposed to the solvent, corroborating with the increased accession of these sites to PK. Additionally, a significant proximity to ion, lipid or ligand binding sites of the enzymes was noted, either by prediction algorithms or manual inspection. Notably, these regions are known to provoke conformational changes⁷ (Figure S6a–e, Table S22).

In addition, considering the large evidence of disulfide bonds and protein glycosylation in previous saliva proteomics data,⁵ and in our own data (Figure S3b, Table S11–S17), we investigated the role of glycosylation on promoting conformational changes on proteins. Although glycopeptides are often present in a lower dynamic range and exhibit reduced ionization efficiency in mass spectrometry compared to nonglycosylated peptides,²⁵ we were able to identify N-glycosylated peptides in our nonenriched samples data, including N-glycans from the main classes: highly truncated, paucimannosidic, oligomannosidic and complex/hybrid N-glycans. To further explore it, we evaluated the Immunoglobulin heavy constant alpha 1 (IGHA1) protein, the only protein in which glycopeptides were identified only in the control samples (Figure 3d–f; Table S23–S26). In our data, two different IGHA1 glycopeptides, one carrying HexNAc(5)Hex(3) and the other carrying HexNAc(4)Hex(5)-Fuc(2)NeuAc(2) N-glycan, were identified. *In silico* modeling suggests a change in conformational upon the presence of glycans, which can also be attributed to a change in the accessibility of the enzyme, although the respective nonglycosylated peptides were not identified in the other groups, even with high protein coverage of about 90% in the control, 91% in pN0 and 89% in pN+. Finally, given that other studies have pointed out an association between the expression of immunoglobulin heavy chains and cancer prognostic and survival prediction,^{26,27} including our results²⁸ showing the same glycosylation site in OSCC tumoral tissues, a deeper look toward their glycosylated state under native conditions could improve even further the patient's classification and its functional role.

CONCLUSIONS

In summary, this conformational comprehensive approach using saliva revealed that the structural readout captured altered proteins and biological processes, suggesting that (i) the integration of these approaches with classical proteomics workflow can improve the detection of novel and robust diagnostic and prognostic-associated proteins, mainly in highly protein dynamic range matrix; (ii) LiP-MS can be a powerful strategy to detect altered biological processes assisting in the search for molecular events underlying physiological and pathological events; and (iii) it allows the prioritization of novel proteins and biological processes associated with OSCC and not revealed by traditional proteomics workflow methods, selecting potential candidates as markers for monitoring in the path of disease or therapeutic targets. This study is a pioneer in the field, showing the diversity of the salivary proteome beyond

the protein abundance associated with oral cancer, opening new possibilities for future studies. Conformatypic proteins and pathways are revealed by LiP-MS associated with OSCC, such as TKT, and IGHA1, which showed changes in protein conformation, probably through protein phosphorylation or glycan composition, respectively. Altogether, our findings show that LiP-MS can identify structural conformational changes in saliva associated with OSCC clinical features and with relevance to the diagnosis and prognosis of the disease.

MATERIALS AND METHODS

Patient Selection

This study comprised two patient cohorts: the first encompassed a pool of saliva samples from five patients with lymph node metastasis (pN+; “p” indicates the presence of metastasis in the lymph node histologically confirmed) and without lymph node metastasis (pN0) for protocol optimization, and an independent patient cohort including healthy control ($n = 4$) and OSCC patients ($n = 8$, pN0 = 4, pN+ = 4). Only patients who had active oral malignant lesions at the time of saliva collection in the presence or absence of lymph node metastasis (pN+ and pN0, respectively) were included.

This study was approved by the Ethics Committee of the Instituto do Câncer do Estado de São Paulo (ICESP), São Paulo, SP, Brazil, through protocol CAAE 30658014.1.1001.0065, and by the Ethics Committee of the Piracicaba Dental School (FOP), São Paulo, SP, Brazil, through protocol CAAE 51762415.5.0000.5418. All participants provided written informed consent. The study was conducted following the Declaration of Helsinki and was performed following the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement (von Elm et al., 2008).²⁹ The procedures used for saliva sampling³⁰ and annotation were performed following the guidelines and experimental protocols approved by both ethics' committees.

Saliva Collection

All saliva samples were collected from individuals who were advised not to eat for at least 1 h, then rinsed their mouths with 5 mL of drinking water, and saliva was subsequently harvested without stimulation into a sterile plastic bottle for 5 min.³⁰ Frozen stocks of saliva were stored at -80°C until further used.³⁰ Before use, saliva cellular debris was first harvested by centrifugation at 1.500g for 5 min at 4°C , and the supernatant was collected and transferred to a fresh micro tube. Protein concentration was determined with the Bradford assay (Bradford Assay Kit, Thermo Fisher Scientific).

Saliva Preparation through Limited Proteolysis (LiP)

For protocol optimization, three distinct protocols with different denaturing agents and concentrations (4 M urea only or with 2 M thiourea, or 0.1% RapiGest) were tested in a pool of healthy individual's saliva samples for protein digestion following LiP-MS. The best protocol (RapiGest) was selected based on the highest number of proteins and peptides identification. Ten micrograms of each proteome extract were subjected to tryptic digestion (as described below) for protein abundance change evaluation (*tryptic digestion-only proteome*). Another 10 μg was submitted for the LiP-MS approach to access protein structural changes, which included a double-protease digestion step with a nonspecific protease followed by digestion with trypsin. First, for the LiP approach, proteinase K (PK) from *Tritirachium album* (Sigma-Aldrich) was added to the samples under native

conditions at an enzyme/substrate (E:S) ratio of 1:100 (w/w) and incubated for 1 min at 25°C with agitation.³¹ In parallel, the same volume of water was added to the *tryptic digestion-only proteome* samples and incubated for 1 min at 25°C with agitation. PK digestion was stopped by heating samples for 5 min at 98°C in a thermocycler, followed by the addition of RapiGest (Waters) to a final concentration of 0.1% for protein denaturation. The same procedure was applied to *tryptic digestion-only proteome* samples. Both the LiP-MS and *tryptic digestion-only proteome* samples were then subjected to complete tryptic digestion in denaturing conditions, as described below.

Tryptic Digestion

The saliva samples analyzed in this study were reduced by incubation with dithiothreitol (DTT) (Sigma-Aldrich) to a final concentration of 5 mM for 25 min at 56°C . Next, the alkylation of free cysteine residues was achieved by adding iodoacetamide-IAA (Sigma-Aldrich) to a final concentration of 14 mM for 30 min at room temperature in the dark. Samples were digested with sequencing-grade porcine trypsin (Promega) at an enzyme/substrate ratio of 1:100, added twice to the samples after 4 h at 37°C , and incubated for 16 h at 37°C . Protease digestion was quenched by lowering the reaction pH (<3) with a final concentration of 0.5% trifluoroacetic acid. The peptide mixtures were loaded onto stage-tip C18 for sample desalting and eluted with 80% acetonitrile in 0.1% formic acid. After elution, the peptides were dried in a vacuum centrifuge, resolubilized in 0.1% formic acid, and analyzed by mass spectrometry. The run order was randomized using an R (v3.4.0) environment to prevent systematic bias during MS acquisition.

LC-MS/MS Data Acquisition

Peptide digests (2 μg) from LiP-MS and *tryptic digestion-only proteome* samples were analyzed on an LTQ Orbitrap Velos (Thermo Fisher Scientific, San Jose, CA) mass spectrometer coupled to nanoflow liquid chromatography on an EASYnLC system (Proxeon Biosystems, Odense, DK) with a Proxeon nanoelectrospray ion source. Peptides were subsequently separated in a 2%–90% acetonitrile gradient in 0.1% formic acid using a PicoFrit analytical column (20 cm $\times$ ID75, 5 μm particle size, New Objective) at a flow rate of 300 nL/min over 220 min, in which a gradient of 35% acetonitrile was reached in 173 min. The nanoelectrospray voltage was set to 2.2 kV, and the source temperature was set to 275°C . The instrument methods employed for LTQ Orbitrap Velos were set up in data-dependent acquisition (DDA) mode. Full-scan MS spectra (m/z 300–1,600) were acquired in the Orbitrap analyzer after accumulation to a target value of 1E^6 . Resolution in the Orbitrap was set to $r = 60,000$, and the 20 most intense peptide ions (top 20) with charge states ≥ 1 were sequentially isolated to a target value of 5,000 and fragmented in the high-pressure linear ion trap by collision-induced dissociation (CID) with a normalized collision energy of 35%. Dynamic exclusion was enabled with an exclusion list size of 500 peptides, an exclusion duration of 60 s, and a repetition count of 1. An activation Q of 0.25 and an activation time of 10 ms were used.

Peptide and Protein Identification

Raw files from both digestion approaches were searched against the Human Uniprot database (version January 2020, 96,525 protein sequences, 38,621,617 residues) using the MaxQuant search engine (version 1.6.20) and PEAKS Studio X (version 7.5, Bioinformatics Solutions Inc.). For the *tryptic digestion-only*

proteome on MaxQuant, up to two missed cleavages were allowed and the cleavage of KP and RP peptide bonds was excluded. For the MaxQuant search of LiP-MS samples, a semispecific tryptic digestion was set, while for *tryptic digestion-only proteome* samples, a fully tryptic digestion was applied. Cysteine carbamidomethylation (+57.0214 Da) and methionine oxidation (+15.99492) were allowed as fixed and variable modifications, respectively; monoisotopic peptide tolerance was set to 10 ppm, and fragment mass tolerance was set to 1 Da for both digestion strategies. The FDR was estimated using the target-decoy method³² and set to 1% at the peptide and protein levels.

For PEAKS analysis, the database combined with *de novo* searches were performed and only peptides identified and quantified in database search output were considered for further analysis. Also, a fully tryptic digestion rule was used for protein abundance quantification for the *tryptic digestion-only proteome*, while using the semispecific tryptic rule for LiP-MS samples. For both types of samples (*tryptic digestion-only proteome* and LiP), PEAKS PTM database (containing most common 32 distinct PTMs provided by the software) were defined and used as variable modifications. The FDR was estimated using the decoy fusion method³³ and set to 1% at the peptide and protein levels. Only peptides confidently identified (PTM Δ score = 20, protein lgP $\geq$ 20) were considered.

Also, Byonic node (Protein Metrics Inc., CA, USA) within Proteome Discoverer v 2.4.0.305 (Thermo Scientific) was used for the identification of intact glycopeptides in both types of samples (*tryptic digestion-only proteome* and LiP), using mammalian glycan database (containing 134 distinct glycans provided by the software) for glycan composition information and Uniprot database (version January 2020, 96,525 protein sequences, 38,621,617 residues) for protein search. For LiP-MS, a semispecific tryptic rule was used for the LiP-MS samples. All searches were filtered to <1% FDR at the protein level and 0% at the peptide level by using a protein decoy database.³⁴ Only N-glycopeptides confidently identified with PEP 2D scores <0.001 were considered.³⁵

LiP-MS Data Analysis

In the LiP-MS samples, only fully and semitryptic peptides that had at least one identified peptide were used for quantification, while in the *tryptic digestion-only proteome* samples only fully tryptic proteotypic peptides were used for protein abundance analysis.

LiP-MS data obtained by all search engines (MaxQuant, PEAKS, and Byonic) were individually normalized at the peptide level by dividing each peptide intensity by the mean intensity value of each sample. Normalized peak areas were used for quantitation in LiP-MS samples, considering fully and semitryptic peptides. Group comparison analysis of peptides (LiP-MS samples) and proteins (*tryptic digestion-only proteome*) was performed independently within Perseus (version 1.6.5) for all search algorithms. At the protein level, LFQ values transformed in Log2 were used for quantification.

Definition and Annotation of Prognostic and Diagnostic Markers

Normalized intensity values for LiP-MS were performed at the peptide level from MaxQuant, and PEAKS searches and were used to determine differentially protein conformation-based changes among control, pN0, and pN+ conditions in Perseus v. 1.6.5 software (ANOVA followed by Tukey's test, p -value <0.05 was set to each data set).⁸ Although p -value correction was

tested, only four significant candidates in Database "only search" and none in Database and *de novo* search were identified after applying Benjamini–Hochberg or Permutation-FDR. Therefore, all further analysis considered significant the p -value not corrected for multiple testing. Glycopeptides identified in the Byonic search were used exclusively for qualitative analysis to access the role of glycans on *in silico* protein conformation, once none of the glycosylated peptides were statistically significant (ANOVA, p -value >0.05) considering the selected criteria. Peptides with significant differential abundances or exclusively detected in one or two out of the three analyzed conditions and present in at least 50% of the samples were grouped as diagnostic or prognostic markers considering their association with OSCC, following the criteria described below:

- (i) Potential diagnostic markers: peptides differentially abundant between control and OSCC (pN0 and/or pN+, ANOVA, p -value <0.05). This category also comprised peptides detected "exclusively" in control or OSCC conditions. Diagnostic markers with higher abundance in OSCC (pN+ and/or pN0) compared to the control or those exclusively detected in OSCC were termed "upregulated," while peptides exhibiting lower expression in OSCC (pN+ and/or pN0) compared to the control or those exclusively detected in the control group were named "downregulated".
- (ii) Potential prognostic markers: peptides differentially abundant between pN+ and pN0 (ANOVA, p -value $\leq$ 0.05) and peptides exclusively identified in pN+ or pN0 conditions. Prognostic markers exhibiting higher expression in pN+ compared to pN0 or those exclusively detected in pN+ were termed "upregulated", while peptides exhibiting lower expression in pN+ compared to pN0 or those exclusively detected in pN0 were termed "downregulated".

Biological Process Enrichment Analysis

For the potential diagnostic and prognostic markers, GO biological processes were enriched using the Molecular Signatures Database from Gene Set Enrichment Analysis software (Fisher's Exact Test, p -value <0.05),^{36,37} and the top-10 overrepresented GO terms were simultaneously visualized for proteins with potential diagnostic and prognostic value, combined with source proteins from diagnostic peptides.

We also tested the proteins with significant abundance changes in *tryptic digestion-only proteome* and the proteins with significant conformational based changes for LiP-MS proteome samples among the control, pN0, and pN+ groups. Overrepresented GO biological processes were determined with Panther^{38,39} tool (Released 2024–02–26; FDR $\leq$ 0.01; Bonferroni correction) considering the GO Ontology database (Released 2024–01–17) and summarized by removing redundant terms in REVIGO.⁴⁰ Molecular pathways were enriched using the ActivePathways⁴¹ tool with hypergeometric tests to determine overrepresented REACTOME terms (FDR $\leq$ 0.01). These terms were used as input for the Enrichment-Map^{42,43} plugin in Cytoscape v.3.9.1⁴⁴ for network visualization and their coloring according to supporting evidence. Enrichment maps for pathways overrepresented in structural and abundance analyses were visualized with a Jaccard and overlap combined coefficient of 0.6. Major biological themes and their relationships were represented as clusters of similar pathways. A p -value threshold of 0.05 was set to determine the significance.

Feature, Thermal Stability, and Icelogo Analysis

To explore which features were enriched among the significantly altered LiP peptides, we used Typic⁴⁵ to retrieve Uniprot database features (e.g., amino acid modifications, PTMs including glycosylation and disulfide bond, mutations) with experimental evidence from potential diagnostic and/or prognostic LiP markers. To verify if protein thermal stability may influence the formation of LiP peptides, we utilized The Meltome Atlas.¹⁴ For this, differential LiP peptides from MaxQuant and PEAKS searches were initially mapped to its respective protein, and then the melting temperature of the proteins was acquired considering the HEK293T data set as a background.

Prioritization of Conformotypic Peptides Associated with OSCC Diagnosis and Prognosis

Prioritization was based on the following criteria: (1) top hit in ranking attribute method analysis; (2) clinical correlation with prognostic features; (3) identifications in previous studies as differentially abundant proteins from nonmalignant cells of primary OSCC tumors compared to committed lymph nodes.²⁰

Three filter approaches implemented in the data mining tool Waikato Environment for Knowledge Analysis (WEKA) v.3.8.5 were applied in the LiP data set. They were (i) information gain attribute evaluation, (ii) chi-square attribute evaluation, and (iii) support vector machine (SVM) attribute evaluation. Also, the ranking method of search, german_credit, is applied to the data set.

To investigate the diagnostic or prognostic value of the significant conformotypic peptides, a linear regression analysis was performed using R code to evaluate the linear relationship between peptide abundance and clinicopathological variables. *P*-value <0.05 was used to define significance. The Pearson-product moment correlation coefficient was calculated to infer the strength of the association between the described variables. Only the peptides that corresponded to peptides associated with patient clinicopathological data (alcohol consumption, smoking habit, sample group, status, lymph node metastasis, and tumor size) with prognostic value were further investigated. Also, peptides were evaluated regarding their distinguishing performance in ranking score.

Structural Analysis

A total of 11 TKT-associated proteins with determined 3D structures were retrieved from the Protein Data Bank (PDB) to identify structural features that may explain the generation of differential conformotypic peptides. A total of 28 conformotypic peptides were analyzed. The PDB structures and the corresponding conformotypic peptides were retrieved. In the case of multiple available structures for a target protein, the PDB with a better resolution was selected. Only PDBs containing the peptide in the structure were considered, and only biounits were included in the analysis.

To assess the secondary structure around the conformotypic peptides, we employed the DSSP software in the Bio3D R package.⁴⁶ The secondary structure was determined within a window of 10 positions N and C-terminal to the conformotypic peptide.

For B-factor analysis, absolute B-factors of amino acids within a window of 10 positions N- and C-terminal to the conformotypic peptide were collected from each PDB structure. The analysis was conducted by using both absolute values and scaled values for each PDB.

To evaluate the solvent accessibility of the region of the conformotypic peptides, we used the DSSP software in the Bio3D R package.⁴⁶ The accessibility was computed with default parameters for each position in a window of 10 positions N- and C-terminal to the target peptides. For this analysis, we used a threshold of 10 Å² to consider a residue exposed to the solvent.

To investigate whether conformotypic peptides are located at putative binding sites of proteins, DNA, or small molecules, we predicted these binding sites using PeSTo.⁴⁷ PeSTo, obtained from <https://github.com/LBM-EPFL/PeSTo>, was locally run on the entire target proteins with the i_v4_1 pretrained model. Following predictions, we analyzed whether the Cα of the N or the C-terminal amino acid of conformotypic peptides is within a distance less than 10 Å from an amino acid with a PeSTo prediction higher than 0.5.

Molecular Dynamic Simulation

To model the *N*-glycosylation of the immunoglobulin heavy constant alpha 1 (IGHA1), the IGHA1 3D structure was obtained from the Protein Data Bank (PDB ID: 6UE7) and was preprocessed to keep only the Ig-like 2 (UNIPROT ID: P01876) domain, which is the target of *N*-glycosylation at position 144. The *N*-glycan residues (HexNAc[4]Hex[5]-Fuc[2]NeuAc[2] and HexNAc[5]Hex[3]) were modeled using the CHARMM-GUI⁴⁸ and Glycan Modeler module.⁴⁹ To access the dynamics of the glycans and their impact on the IGHA1 structure, we used three initial models: (1) the IGHA1 Ig-like 2 domain without glycan, (2) the HexNAc(4) Hex(5)-Fuc(2)NeuAc(2) *N*-glycan, or (3) the HexNAc(5)Hex(3) *N*-glycan. For MD simulations, the N-terminal and C-terminal caps were added using CHARMM-GUI and the Solution Builder module was used to generate a rectangular solvation box (edge distance of 10 Å from the solute) filled with TIP3P water. Na⁺ and Cl[−] counterions were added to reach net-neutralization with salt excess to reach 150 mM NaCl. FF14SB and GLYCAM-06j force fields were used for protein and glycan residues, respectively. All MD simulations were performed using the Amber20 suite. The system was minimized by 2500 steps of steepest descent minimization and 2500 steps of the conjugate gradient. Equilibration was performed using the NVT ensemble (200 ps), followed by the NPT ensemble (200 ps), followed by the harmonic restraints in solute atoms. A final NPT equilibration step without restraints was performed for 500 ps. For each glycoprotein system and the IGHA1 system without glycans, the production run was performed at 298 K for 100 ns with a time step of 2 fs in triplicate. Hydrogen-containing bonds were constrained using SHAKE.⁵⁰ Long-range electrostatic interactions were calculated using particle mesh Ewald, and short-range nonbonded interactions were calculated with a 9 Å cutoff.

To analyze whether the glycans could affect the IGHA1 structure, the distance between each IGHA1 residue (only considering Cα) along the simulation trajectory and its position in the reference structure was computed. The first frame of the IGHA1 simulation trajectory without glycan (WT) was used as a structural reference. Then, the median of the average deviations in each triplicate in the simulations with glycan was subtracted from the median of the deviation without glycans (WT). The deviations were computed by considering only the last half of the trajectory. All analyses were performed using VMD⁵¹ and the R package Bio3D.⁴⁶

Data Visualization and Processing

Heatmap diagrams, bar plots, and regression model plots were created using data visualization in the R (version 3.6.0) environment and were also based on identified modified peptide sequences. Amino acid frequency and reconstruction of cleavage site was performed as previously described.¹¹ Common identifications between prognostic and diagnostic proteins were visualized in Venn diagrams using the InteractiVenn tool.⁵² To visualize the prioritized target, TKT, and its conformational peptides in a 3D view, was analyzed in Chimera X using pdb data.

Experimental workflow and graphical abstract were visualized using BioRender (BioRender.com).

Materials Availability

This study did not generate new unique reagents.

■ ASSOCIATED CONTENT

Data Availability Statement

The mass spectrometry proteomic data have been deposited in the ProteomeXchange Consortium (<http://proteomecentral.proteomexchange.org>) via the PRIDE partner repository (Vizcaino et al., 2013) with the data set identifier PXD042574. All other data supporting the findings of this study are available from the corresponding author on request.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jproteome.4c00093>.

Overview of the experimental design (Figure S1); Quality control analysis of LC-MS/MS runs (Figure S2); Overview of LiP differential proteins and sites (Figure S3); Structural and abundance changes between patient groups (Figure S4); Target prioritization (Figure S5); Analysis of structure-based features (Figure S6) (PDF)

Clinical and pathological information, Randomized order of LC-MS/MS sample acquisition (Table S1 and S2); Tryptic proteome and LiP proteome (Tables S3–S8); Differentially abundant LiP peptides (Tables S9–S10); GO biological processes enrichment (Tables S11–S13); Clinicopathological correlation (Table S14); PTMs evaluation (Tables S15–S21); LiP glycopeptides and nonglycopeptides identified, quantified and selected using the “Byonic search (Byos software)” (Tables S22–S25); Conformational LiP peptides in TKT network (Table S26) (XLSX)

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D.C.G. and A.F.P.L. designed research; A.R.S.S., M.A.L., T.B.B., and A.C.P.R. collected, processed, and stratified clinical samples; D.C.G., S.Y., and A.G.C.N were responsible for saliva collection,

performed LiP-MS sample preparation and LiP-MS screens; B.P. and R.R.D. performed LC-MS/MS data acquisition and mass spectrometer maintenance; D.C.G., L.D.T., C.M.C., A.F.B.L., G.P.T., H.V.R.F., F.M.S.P., and G.A.C. analyzed and interpreted data and led bioinformatics analysis across the project; D.C.G., A.G.C.N., L.D.T., C.M.C., A.F.B.L., and A.F.P.L. wrote the manuscript with contributions from all authors. D.C.G. and A.F.P.L. edited the manuscript and coordinated revision.

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Notes

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■ ABBREVIATIONS

OSCC, oral squamous cell carcinoma; pN0, pathologically confirmed absence of lymph node metastasis; pN+, pathologically confirmed presence of lymph node metastasis; TNM, tumor (T), nodes (N), metastasis (M) classification system; LC-MS/MS, liquid chromatography–tandem mass spectrometry; FDR, false discovery rate; LiP, limited proteolysis; LiP-MS, limited proteolysis–mass spectrometry analysis; DDA, data dependent analysis.

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